



COMMAND REFERENCE

VIRTUAL CONTROL
VMWARE CLOUD FOUNDATION SOLUTIONS

VMware Cloud Foundation 9.0 Command Reference

Comprehensive command reference for VCF 9.0 operations covering ESXi, vSAN, networking, certificates, Cloud Builder, SDDC Manager, and Linux administration.

VCF 9.0

COMMAND REFERENCE

ESXI

VSAN

VCF 9.0

VMware Cloud Foundation

VMware Cloud Foundation 9.0 Command Reference

Comprehensive Command Guide Extracted from VCF 9.0 Documentation

Where commands run in this reference. Despite "VCF" in the name, this reference is **ESXi-host-focused** — almost every command runs from an **ESXi host shell (SSH to host root)**. Commands include `esxcli`, `vmkfstools`, `vim-cmd`, `localcli`, `vsish`. A small set are workstation-side PowerCLI equivalents (clearly labeled). Replace any credential placeholder like `<your-password>` with values from your vault.

Version: 1.1 **Date:** May 2026 **Source:** VCF 9.0 Troubleshooting Workbook, Test Questions & Lab Documentation

PowerShell / PowerCLI Translation Notice
Most commands in this reference run on the ESXi host shell. Operators working from a Windows jump host can use PowerCLI equivalents for commands that map to vSphere API operations. See the central [PowerShell Translation Reference](#) for cmdlet mappings, authentication patterns, and the one-time PowerCLI configuration. Commands flagged # Run on ESXi host shell only have no PowerCLI equivalent and must be executed via SSH on the host.

One-time PowerCLI configuration (run once per workstation):

```
Set-PowerCLIConfiguration -InvalidCertificateAction Ignore -ParticipateInCEIP $false -Confirm:$false  
| Out-Null
```

Revision History

Version	Date	Author	Changes
1.0	January 2026	Virtual Control LLC	Initial release.
1.1	May 2026	Virtual Control LLC	Added PowerShell/PowerCLI translation callouts and equivalents for ESXi shell commands; added link to central translation reference; flagged ESXi-shell-only blocks.

Table of Contents

1. [ESXi Host Administration](#)
2. [ESXi Storage Management](#)
3. [vSAN Configuration](#)
4. [ESXi Network Commands](#)
5. [ESXi Services Management](#)
6. [Certificate Management](#)
7. [Cloud Builder / VCF Installer Commands](#)
8. [SDDC Manager Commands](#)
9. [Linux Administration](#)
10. [Network Troubleshooting](#)
11. [VMware Workstation Configuration](#)

1. ESXi Host Administration

1.1 Hostname Configuration

Command	Description
<code>esxcli system hostname get</code>	Displays the current hostname configuration including FQDN, short name, and domain name configured on the ESXi host
<code>esxcli system hostname set --fqdn=hostname.domain.local</code>	Sets the fully qualified domain name (FQDN) for the ESXi host. This affects SSL certificate generation and how the host identifies itself

Detailed Usage:

```
# bash on ESXi host shell only
# Check current hostname configuration
esxcli system hostname get

# Output example:
#   Domain Name: lab.local
#   Fully Qualified Domain Name: esxi01.lab.local
#   Host Name: esxi01

# Set new FQDN (use before certificate regeneration)
esxcli system hostname set --fqdn=esxi01.lab.local
```

PowerShell/PowerCLI equivalent (run on Windows workstation with PowerCLI):

```
# Connect to vCenter (or directly to host)
Connect-VIServer -Server vcenter.lab.local

$h = Get-VMHost -Name 'esxi01.lab.local'
$esx = Get-EsxCLI -V2 -VMHost $h

# Get hostname configuration
$esx.system.hostname.get.Invoke()

# Set FQDN (parameter spec via CreateArgs ensures correct binding)
$args = $esx.system.hostname.set.CreateArgs()
$args.fqdn = 'esxi01.lab.local'
$esx.system.hostname.set.Invoke($args)
```

1.2 SSL Certificate Management

Command	Description
<code>/sbin/generate-certificates</code>	Regenerates the default ESXi SSL certificates using the current hostname settings. Run this after changing the FQDN to update certificates with correct hostname
<code>openssl x509 -in /etc/vmware/ssl/rui.crt -text -noout</code>	Displays the full details of the ESXi SSL certificate including subject, issuer, validity dates, and Subject Alternative Names (SAN)
<code>openssl x509 -in /etc/vmware/ssl/rui.crt -text -noout grep -A1 "Subject Alternative Name"</code>	Shows only the Subject Alternative Name field from the certificate - useful for verifying the hostname in the certificate

Certificate File Locations: - Certificate: `/etc/vmware/ssl/rui.crt` - Private Key: `/etc/vmware/ssl/rui.key`

Detailed Usage:

```
# Run on ESXi host shell only – generate-certificates and services.sh have no PowerCLI equivalent
# Step 1: Check current certificate SAN
openssl x509 -in /etc/vmware/ssl/rui.crt -text -noout | grep -A1 "Subject Alternative Name"

# Step 2: If incorrect, set correct FQDN
esxcli system hostname set --fqdn=esxi01.lab.local

# Step 3: Regenerate certificates
/sbin/generate-certificates

# Step 4: Restart management services
services.sh restart

# Step 5: Verify new certificate
openssl x509 -in /etc/vmware/ssl/rui.crt -text -noout | grep -A1 "Subject Alternative Name"
```

PowerShell/PowerCLI equivalent (run on Windows workstation with PowerCLI):

```
# Hostname can be set via Get-ESXCLI; certificate regeneration and services.sh
# restart still require an SSH session to the host (no PowerCLI cmdlet).
# For vCenter-managed CA-signed certs, see VMCA workflows in vCenter UI.

$h = Get-VMHost -Name 'esxi01.lab.local'
$esx = Get-ESXCLI -V2 -VMHost $h

# Step 2 equivalent: set FQDN
$a = $esx.system.hostname.set.CreateArgs(); $a.fqdn = 'esxi01.lab.local'
$esx.system.hostname.set.Invoke($a)

# Steps 1, 3, 4, 5 require SSH (Posh-SSH module example):
# Install-Module Posh-SSH -Scope CurrentUser
$cred = Get-Credential -UserName 'root' -Message 'ESXi root'
$s = New-SSHSession -ComputerName 'esxi01.lab.local' -Credential $cred -AcceptKey
Invoke-SSHCommand -SessionId $s.SessionId -Command 'openssl x509 -in /etc/vmware/ssl/rui.crt -text -noout | grep -A1 "Subject Alternative Name"'
Invoke-SSHCommand -SessionId $s.SessionId -Command '/sbin/generate-certificates'
Invoke-SSHCommand -SessionId $s.SessionId -Command 'services.sh restart'
Remove-SSHSession -SessionId $s.SessionId
```

1.3 Service Management

Command	Description
<code>services.sh restart</code>	Restarts ALL ESXi management services including hostd, vpxa, and others. Use after certificate changes or configuration updates
<code>/etc/init.d/hostd restart</code>	Restarts only the hostd management service. Less disruptive than full services restart
<code>/etc/init.d/SSH status</code>	Checks the current status of the SSH service (TSM-SSH)
<code>/etc/init.d/SSH start</code>	Starts the SSH service to enable remote command-line access
<code>/etc/init.d/SSH stop</code>	Stops the SSH service to disable remote command-line access

ESXi Service Control:

```
# bash on ESXi host shell only
# List all services
/etc/init.d/ [TAB][TAB] # Shows available services

# Common services:
# - hostd: ESXi management daemon
# - vpxa: vCenter agent
# - SSH: Secure Shell access
# - ntpd: Network Time Protocol
```

PowerShell/PowerCLI equivalent (run on Windows workstation with PowerCLI):

```
# List ESXi host services and their state via vSphere API
$h = Get-VMHost -Name 'esxi01.lab.local'
Get-VMHostService -VMHost $h | Select-Object Key, Label, Running, Policy

# Restart a specific service (e.g. SSH / TSM-SSH)
Get-VMHostService -VMHost $h | Where-Object { $_.Key -eq 'TSM-SSH' } | Restart-VMHostService -
Confirm:$false

# Note: services.sh restart (full management stack restart) has no PowerCLI cmdlet
# and must be run via SSH on the host.
```

1.4 Time Configuration

Command	Description
<code>esxcli system time get</code>	Displays the current system time on the ESXi host
<code>esxcli network firewall ruleset set -e true -r ntpClient</code>	Enables the NTP client firewall ruleset to allow outbound NTP traffic for time synchronization

2. ESXi Storage Management

2.1 Device Listing and Information

Command	Description
<code>esxcli storage core device list</code>	Lists all storage devices connected to the host with detailed information including capacity, vendor, model, and SSD status
<code>esxcli storage core device list \ grep -E "Display Name\ Is SSD"</code>	Filters device list to show only device names and SSD status - useful for verifying disk type recognition
<code>esxcli storage core device list \ grep -A5 "Is SSD"</code>	Shows the SSD field and 5 lines following it for context about each device
<code>esxcli storage core adapter rescan --all</code>	Rescans all storage adapters to detect newly added or changed storage devices
<code>esxcli storage filesystem list</code>	Lists all mounted filesystems including VMFS datastores and their capacity

Example Output Analysis:

```
# bash on ESXi host shell only
# Check if disks are recognized as SSD
esxcli storage core device list | grep -E "Display Name|Is SSD"

# Expected output for SSD:
#   Display Name: Local SATA Disk (...)
#   Is SSD: true

# Problematic output (needs VMX fix):
#   Display Name: Local SATA Disk (...)
#   Is SSD: false
```

PowerShell/PowerCLI equivalent (run on Windows workstation with PowerCLI):

```
$h = Get-VMHost -Name 'esxi01.lab.local'
$esx = Get-EsxCLI -V2 -VMHost $h

# Pull device list via esxcli over the API and project the SSD flag
$esx.storage.core.device.list.Invoke() |
    Select-Object DisplayName, Device, IsSSD, Size, Vendor, Model |
    Format-Table -AutoSize

# Filter to only flag false-SSD detection (lab/Workstation nested VMs)
$esx.storage.core.device.list.Invoke() |
    Where-Object { $_.IsSSD -eq $false } |
    Select-Object DisplayName, Device, IsSSD
```

2.2 Partition Management

Command	Description
<code>partedUtil getptbl /dev/disks/[device-name]</code>	Displays the partition table of the specified disk including partition numbers, start/end sectors, and types
<code>partedUtil mklabel /dev/disks/[device-name] gpt</code>	Creates a fresh GPT (GUID Partition Table) label on the disk, effectively removing ALL existing partitions. DESTRUCTIVE - erases all data

Detailed Usage:

```
# Run on ESXi host shell only – partedUtil has no PowerCLI equivalent
# Step 1: List all disk devices to find device names
esxcli storage core device list

# Step 2: Check existing partitions on a disk
partedUtil getptbl /dev/disks/naa.600508b4000156d7000100000003000

# Step 3: Clear disk and create fresh GPT label
partedUtil mklabel /dev/disks/naa.600508b4000156d7000100000003000 gpt

# Step 4: Verify disk is now empty
partedUtil getptbl /dev/disks/naa.600508b4000156d7000100000003000
```

partedUtil is an ESXi-shell-only utility with no vSphere API surface; it must be run via SSH on the host. The PowerCLI workflow for the same outcome is to remove the disk from any vSAN disk group (`Remove-VsanDiskGroup` / `Get-VsanDisk` | `Remove-VsanDisk`), which clears vSAN partitioning. For full GPT wipe, SSH to the host.

Warning: The mklabel command destroys all data on the disk. Ensure you target the correct device.

3. vSAN Configuration

3.1 Cluster Management

Command	Description
<code>esxcli vsan cluster leave</code>	Forces the ESXi host to leave its current vSAN cluster. Use when you need to remove a host from vSAN or reset its cluster membership
<code>esxcli vsan cluster get</code>	Displays current vSAN cluster information and membership status

3.2 Storage Auto-Mode

Command	Description
<code>esxcli vsan storage automode set --enabled=false</code>	Disables vSAN automatic disk claiming. Prevents vSAN from automatically claiming eligible disks, allowing manual disk selection for cache and capacity tiers
<code>esxcli vsan storage automode set --enabled=true</code>	Enables vSAN automatic disk claiming (default behavior)
<code>esxcli vsan storage automode get</code>	Shows current auto-mode status

When to Disable Auto-Mode: - Before manually configuring vSAN disk groups - When troubleshooting disk claiming issues - In lab environments where you want precise control over disk assignment

4. ESXi Network Commands

4.1 vSwitch Management

Command	Description
<code>esxcli network vswitch standard list</code>	Lists all standard virtual switches with their configuration including uplinks, port groups, and MTU settings
<code>esxcli network vswitch dvs vmware list</code>	Lists distributed virtual switch configurations

4.2 Firewall Management

Command	Description
<code>esxcli network firewall ruleset list</code>	Lists all firewall rulesets and their enabled/disabled status
<code>esxcli network firewall ruleset list \ grep -i ssh</code>	Filters firewall rules to show SSH-related rulesets
<code>esxcli network firewall ruleset set -e true -r sshServer</code>	Enables the SSH server firewall ruleset

4.3 Connection Monitoring

Command	Description
<code>esxcli network ip connection list</code>	Lists all active network connections on the host
<code>esxcli network ip connection list \ grep [port]</code>	Filters connections by a specific port number

5. ESXi Services Management

5.1 SSH Service Control

Command	Description
TSM-SSH	The ESXi service name that controls SSH access (Tech Support Mode SSH)
<code>vim-cmd hostsvc/enable_ssh</code>	Enables SSH service using vim-cmd (alternative method)
<code>vim-cmd hostsvc/disable_ssh</code>	Disables SSH service using vim-cmd

Enabling SSH via ESXi Host Client: 1. Navigate to: Host > Manage > Services 2. Find "TSM-SSH" 3. Click "Start" 4. Set Policy: "Start and stop with host" for persistence

6. Certificate Management

6.1 OpenSSL Certificate Generation

Command	Description
<code>openssl req -x509 -newkey rsa:2048 -keyout server.key -out server.crt -days 365 -nodes -subj "/CN=[hostname]"</code>	Generates a self-signed certificate and private key. The -nodes flag means no passphrase on the private key

Complete Certificate Generation with SANs:

```
# bash on workstation (Linux/macOS/WSL with openssl)
openssl req -x509 -newkey rsa:2048 \
  -keyout server.key \
  -out server.crt \
  -days 365 \
  -nodes \
  -subj "/CN=192.168.1.52/O=VCF-Depot/C=US" \
  -addext "subjectAltName=IP:192.168.1.52,DNS:localhost" \
  -addext "keyUsage=digitalSignature,keyEncipherment" \
  -addext "extendedKeyUsage=serverAuth"
```

PowerShell equivalent (run on Windows workstation — pure PowerShell, no PowerCLI required):

```
# Native Windows self-signed cert with SANs via New-SelfSignedCertificate
$cert = New-SelfSignedCertificate `
  -Subject 'CN=192.168.1.52, O=VCF-Depot, C=US' `
  -TextExtension @(
    '2.5.29.17={text}IPAddress=192.168.1.52&DNS=localhost',
    '2.5.29.37={text}1.3.6.1.5.5.7.3.1',
    '2.5.29.15={critical}{hex}a0'
  ) `
  -KeyAlgorithm RSA -KeyLength 2048 `
  -NotAfter (Get-Date).AddDays(365) `
  -CertStoreLocation 'Cert:\CurrentUser\My'

# Export public cert (.crt) and private key (.pfx -> .key conversion needs openssl)
Export-Certificate -Cert $cert -FilePath .\server.crt -Type CERT | Out-Null
$pwd = ConvertTo-SecureString -String 'temp' -Force -AsPlainText
Export-PfxCertificate -Cert $cert -FilePath .\server.pfx -Password $pwd | Out-Null

# Note: extracting an unencrypted RSA private key in PEM format from the PFX
# still requires openssl: `openssl pkcs12 -in server.pfx -nocerts -nodes -out server.key`
# If you need a Linux-style key file, the openssl bash command above is simpler.
```

Parameter Breakdown: | Parameter | Description | |-----|-----| | -x509 | Generate self-signed certificate instead of CSR | | -newkey rsa:2048 | Create new 2048-bit RSA key | | -keyout | Output file for private key | | -out | Output file for certificate | | -days 365 | Certificate validity period | | -nodes | No DES encryption on private key (no passphrase) | | -subj '/CN=...' | Subject line - CN is Common Name. **Format requires leading slash: /CN=hostname** | | -addext | Add X.509 extensions |

6.2 Java Keystore Management (keytool)

Command	Description
<code>keytool -import -trustcacerts -alias [name] -file [cert] -keystore [cacerts] -storepass changeit -noprompt</code>	Imports a certificate into a Java truststore. The -noprompt flag accepts the certificate without confirmation
<code>keytool -list -keystore [cacerts] -storepass changeit</code>	Lists all certificates in a Java keystore
<code>keytool -delete -alias [name] -keystore [cacerts] -storepass changeit</code>	Removes a certificate from a Java keystore

Default Java Keystore Password: `changeit` (industry standard default)

Keytool Parameters: | Parameter | Description | |-----|-----| | `-import` | Import a certificate into the keystore | | `-trustcacerts` | Trust certificates in the cacerts keystore | | `-alias` | Name/alias for the certificate entry | | `-file` | Path to the certificate file to import | | `-keystore` | Path to the Java keystore file | | `-storepass` | Password for the keystore (default: changeit) | | `-noprompt` | Don't prompt for confirmation |

Cloud Builder Certificate Import Example:

```
# bash on Cloud Builder / VCF Installer appliance (SSH as root)
# Find Java cacerts location
find / -name "cacerts" 2>/dev/null

# Import certificate to Cloud Builder truststore
keytool -import -trustcacerts -alias vcf-depot \
  -file /tmp/depot.crt \
  -keystore /usr/lib/jvm/openjdk-java17-headless.x86_64/lib/security/cacerts \
  -storepass changeit -noprompt
```

PowerShell equivalent (run on Windows workstation — drives the Cloud Builder appliance via SSH):

```
# keytool runs inside the Cloud Builder appliance JVM; there is no PowerCLI cmdlet.
# Drive the import remotely via Posh-SSH from your workstation.

# Install-Module Posh-SSH -Scope CurrentUser # one-time
$cred = Get-Credential -UserName 'root' -Message 'Cloud Builder root'
$cb = New-SSHSession -ComputerName '192.168.1.x' -Credential $cred -AcceptKey

# Push the certificate up
Set-SCPItem -ComputerName '192.168.1.x' -Credential $cred -Path .\depot.crt -Destination '/tmp/'

# Locate cacerts and import
Invoke-SSHCommand -SessionId $cb.SessionId -Command 'find / -name cacerts 2>/dev/null'
$cmd = "keytool -import -trustcacerts -alias vcf-depot -file /tmp/depot.crt " +
  "-keystore /usr/lib/jvm/openjdk-java17-headless.x86_64/lib/security/cacerts " +
  "-storepass changeit -noprompt"
Invoke-SSHCommand -SessionId $cb.SessionId -Command $cmd

Remove-SSHSession -SessionId $cb.SessionId
```

7. Cloud Builder / VCF Installer Commands

7.1 Service Management

Command	Description
<code>systemctl list-units --type=service \\\n grep vcf</code>	Lists all systemd services containing 'vcf' - shows VCF-related services and their status
<code>systemctl restart lcm.service</code>	Restarts the Lifecycle Manager service. Required after certificate imports to pick up new trusted certificates
<code>systemctl restart sddc-manager-ui-app.service</code>	Restarts the SDDC Manager UI application service
<code>systemctl status lcm.service</code>	Shows the status of the LCM service

7.2 Disk Space Management

Command	Description
<code>df -h</code>	Displays filesystem disk space usage in human-readable format (KB, MB, GB)
<code>df -h /tmp</code>	Shows disk space for the /tmp filesystem specifically
<code>rm -rf /tmp/*</code>	Removes all files in /tmp directory. Use when /tmp is full causing trust store creation failures
<code>du -sh /path/*</code>	Shows disk usage summary for each item in a directory

Cloud Builder Storage Locations: | Directory | Purpose | Typical Size | |-----|-----|-----| | /tmp | Temporary files, trust stores | Small (tmpfs) | | /nfs/vmware/vcf/nfs-mount | Bundle storage, large downloads | 400+ GB | | /nfs/vmware/vcf/nfs-mount/bundle | VCF bundle depot | Large |

7.3 File Operations

Command	Description
<code>find / -name "cacerts" 2>/dev/null</code>	Searches entire filesystem for Java cacerts file, suppressing permission errors
<code>find / -name '*.ova'</code>	Finds all .ova files on the system - useful for locating downloaded appliances
<code>cat /etc/resolv.conf</code>	Displays DNS resolver configuration including nameservers and search domains

8. SDDC Manager Commands

8.1 Connection and Status

Command	Description
<code>ssh admin@<sddc-manager-ip></code>	SSH into SDDC Manager as admin user
<code>systemctl status vcf-services</code>	Shows status of all VCF services on SDDC Manager
<code>systemctl restart vcf-services</code>	Restarts all VCF services
<code>systemctl status domainmanager</code>	Shows Domain Manager service status
<code>systemctl status operationsmanager</code>	Shows Operations Manager service status
<code>systemctl status nginx</code>	Shows NGINX web server status
<code>systemctl status postgresql</code>	Shows PostgreSQL database status

8.2 Service Control

Command	Description
<code>systemctl restart domainmanager</code>	Restarts the Domain Manager service
<code>systemctl restart lcm</code>	Restarts Lifecycle Manager
<code>systemctl restart operationsmanager</code>	Restarts Operations Manager
<code>systemctl restart nginx</code>	Restarts the web server
<code>systemctl restart postgresql</code>	Restarts the database

8.3 API Interaction

Command	Description
<code>curl -k -X POST https://localhost/v1/tokens -H "Content-Type: application/json" -d '{"username":"admin","password":"xxx"}'</code>	Obtains API authentication token
<code>curl -k -X PATCH https://localhost/v1/tasks/<task-id> -H "Authorization: Bearer <token>"</code>	Updates a task via API

9. Linux Administration

9.1 Time Synchronization

Command	Description
<code>ntpdate -s [ntp-server]</code>	Forces immediate time synchronization with specified NTP server. The -s flag makes it silent (no output)
<code>date</code>	Displays current system date and time

Command	Description
<code>timedatectl</code>	Shows comprehensive time configuration including NTP status

Example:

```
# bash on Linux appliance (Cloud Builder / SDDC Manager / VCF Operations) via SSH
# Sync time with NTP server
ntpdate -s 192.168.1.230

# Verify time is correct
date
```

PowerShell equivalent (run on Windows workstation — for ESXi host time sync via vSphere API):

```
# For ESXi host NTP configuration – managed via vSphere API
$h = Get-VMHost -Name 'esxi01.lab.local'

# View NTP servers and service state
Get-VMHostNtpServer -VMHost $h
Get-VMHostService -VMHost $h | Where-Object { $_.Key -eq 'ntpd' }

# Add an NTP server and start the daemon
Add-VMHostNtpServer -VMHost $h -NtpServer '192.168.1.230'
Get-VMHostService -VMHost $h | Where-Object { $_.Key -eq 'ntpd' } |
    Start-VMHostService -Confirm:$false
Get-VMHostService -VMHost $h | Where-Object { $_.Key -eq 'ntpd' } |
    Set-VMHostService -Policy 'On' -Confirm:$false

# For Linux appliances (Cloud Builder, SDDC Manager) – drive via Posh-SSH
# (ntpdate / timedatectl have no PowerCLI surface):
# Invoke-SSHCommand -SessionId $s.SessionId -Command 'ntpdate -s 192.168.1.230'
```

9.2 DNS Resolution

Command	Description
<code>nslookup [hostname]</code>	Queries DNS servers to resolve a hostname to IP address - tests DNS resolution
<code>cat /etc/resolv.conf</code>	Shows DNS configuration file with nameservers and search domains

9.3 File Downloads

Command	Description
<code>curl -k -u user:pass URL</code>	Downloads a file using basic authentication. -k ignores SSL certificate verification, -u specifies username:password
<code>curl -k -O URL</code>	Downloads file and saves with original filename
<code>curl -I URL</code>	Fetches only HTTP headers (useful for testing connectivity)
<code>wget URL</code>	Alternative download command

curl Parameters: | Parameter | Description | |-----|-----| | `-k` | Ignore SSL certificate errors (insecure) | | `-u user:pass` | HTTP Basic Authentication | | `-o` | Save with original filename | | `-o filename` | Save with specified filename | | `-I` | HEAD request only (headers) |

9.4 SSH Options

Command	Description
<code>ssh -o StrictHostKeyChecking=no root@[ip] "command"</code>	SSH with host key verification disabled - useful for automation but reduces security
<code>scp file.ova root@[ip]:/path/</code>	Secure copy file to remote host

SSH Option: StrictHostKeyChecking=no - Disables the prompt for unknown host keys - Use cautiously - only in trusted lab environments - Alternative: `UserKnownHostsFile=/dev/null` to not save keys

10. Network Troubleshooting

10.1 Port Testing

Command	Description
<code>nc -zv [ip] [port]</code>	Tests TCP port connectivity using netcat. <code>-z</code> scans without sending data, <code>-v</code> provides verbose output
<code>nc -zv [ip] 22</code>	Tests if SSH port (22) is open on remote host
<code>nc -zv [ip] 443 -w 5</code>	Tests HTTPS port with 5-second timeout

Common Ports to Test: | Port | Service | |-----|-----| | 22 | SSH | | 443 | HTTPS | | 8443 | Custom HTTPS (offline depot) | | 902 | VMware authentication |

10.2 Connectivity Testing

Command	Description
<code>ping [ip/hostname]</code>	Tests basic network connectivity
<code>curl -I https://[server]</code>	Tests HTTPS connectivity and retrieves headers

11. VMware Workstation Configuration

11.1 VMX File Settings for Virtual SSDs

Virtual disks in VMware Workstation default to HDD emulation. To mark them as SSDs for vSAN:

Setting	Description
<code>sata0:0.virtualSSD = 1</code>	Marks SATA controller 0, disk 0 as SSD
<code>sata0:3.virtualSSD = 1</code>	Marks SATA controller 0, disk 3 as SSD
<code>scsi0:0.virtualSSD = 1</code>	Marks SCSI controller 0, disk 0 as SSD
<code>nvme0:0.virtualSSD = 1</code>	Marks NVMe controller 0, disk 0 as SSD

Format: `[controller]:[disk].virtualSSD = 1`

VMX Edit Process: 1. Power off the VM in VMware Workstation 2. Locate and edit the VMX file (e.g., `esxi01.vmx`) 3. Add the virtualSSD lines for each disk 4. If VM is in library: Remove from Library, then File > Open VMX 5. Power on and verify with: `esxcli storage core device list | grep "Is SSD"`

Quick Reference Card

Most Common Commands

Task	Command
Set ESXi hostname	<code>esxcli system hostname set --fqdn=host.domain.local</code>
Regenerate ESXi certs	<code>/sbin/generate-certificates</code>
Restart ESXi services	<code>services.sh restart</code>
Check disk SSD status	<code>esxcli storage core device list \ grep "Is SSD"</code>
Clear disk partitions	<code>partedUtil mklabel /dev/disks/[device] gpt</code>
Disable vSAN auto-claim	<code>esxcli vsan storage automode set --enabled=false</code>
Leave vSAN cluster	<code>esxcli vsan cluster leave</code>
Import Java certificate	<code>keytool -import -trustcacerts -alias [name] -file [cert] -keystore [cacerts] -storepass changeit -noprompt</code>
Restart Cloud Builder LCM	<code>systemctl restart lcm.service</code>
Check disk space	<code>df -h</code>
Clean /tmp	<code>rm -rf /tmp/*</code>
Test SSH port	<code>nc -zv [ip] 22</code>

Task	Command
Sync NTP time	<code>ntpdate -s [ntp-server]</code>
Check DNS resolution	<code>nslookup [hostname]</code>

Default Credentials Reference

Component	Default Password	Notes
Cloud Builder SSH (root)	vmware	Change in production
Java Keystore	changeit	Industry standard

Important File Locations

File	Purpose
<code>/etc/vmware/ssl/rui.crt</code>	ESXi SSL certificate
<code>/etc/vmware/ssl/rui.key</code>	ESXi SSL private key
<code>/etc/resolv.conf</code>	DNS configuration
<code>/usr/lib/jvm/.../lib/security/cacerts</code>	Java truststore
<code>/nfs/vmware/vcf/nfs-mount/bundle</code>	VCF bundle depot

Document End

Compiled from VCF 9.0 Troubleshooting documentation - January 2026